Oppgaver for kapittel 0

0.2.1

Løs likningene.

a) $x + 8 = 18$ b) $x - 3 = 2$ c) $x - 8 = 1$

d) $x + 12 = 14$ e) $x - 1 = 2$ f) $x - 3 = 1$

g) $21 = x + 11$ h) $24 = x + 16$ i) $4 = x - 6$

0.2.2

Løs ligningene.

a) $16x - 20 = 15x + 17$ b) $18x - 11 = 17x + 18$

c) $17x - 15 = 16x + 8$ d) $4x - 9 = 6 + 3x$

e) $12x - 6 = 11x + 2$ f) $2x + 10 = 3x - 1$

g) $5 + 8x = 9x - 18$ h) $15 + 2x = 3x - 4$

i) $9x + 8 = 10x - 2$ j) $17x + 9 = 18x - 19$

0.2.3

Løs ligningene.

a) $3x = 12$ b) $10x = 50$ c) $7x = 63$ d) $2x = 30$

0.2.4

Løs ligningene.

a) $\dfrac{x}{4} = 2$ b) $\dfrac{x}{9} = 8$ c) $\dfrac{x}{7} = 7$ d) $\dfrac{x}{15} = 10$

0.2.5

Løs ligningene.

a) $18x - 27 = 9x + 36$ b) $7x - 27 = 4x + 3$

c) $15x - 16 = 7x + 32$ d) $13x - 42 = 7x + 12$

e) $4 + 9x = 13x - 32$ f) $7x + 8 = 11x - 24$

g) $5x + 4 = 8x - 11$ h) $7 + 10x = 14x - 9$

0.2.6

Løs ligningene.

a) $\dfrac{x}{4} + \dfrac{3}{2} = 4$ b) $\dfrac{15}{6} = \dfrac{x}{3} - \dfrac{3}{2}$

c) $\dfrac{5}{3} - \dfrac{x}{2} = -7$ d) $\dfrac{x}{2} + \dfrac{4}{3} = \dfrac{9}{5}$

0.2.7

Gitt en rettvinklet trekant $\triangle ABC$, hvor $\angle C = 90°$. Vis at

$$\angle A = 90° - \angle B$$

0.2.8 (E22)

Løs ligningen

$$3 \cdot 24 \cdot 9 = 4 \cdot 9 \cdot x$$

0.5.1 (1TV21D1)

Løs ligningssystemet

$$2x - y = 4$$

$$x - 2y = 5$$

Gruble 1

Løs ligningene

a) $x - \frac{x}{6} = \frac{x+2}{2}$ b) $\frac{20-x}{3} + 4 = \frac{x+48}{5}$

c) $-\frac{5+x}{7} = -2 + \frac{9-x}{6}$ d) $\frac{2}{3}(x-5) = x - 7$

Gruble 2

Løs ligningene

a) $3 + \frac{5x}{7} = \frac{x}{3} - 1$ b) $-2 - \frac{4x}{9} = \frac{5x}{6} - 7$

c) $x - \frac{2x}{3} = \frac{x-9}{2}$ d) $\frac{x-3}{2} + \frac{4-x}{7} = \frac{1-4x}{2} + 5$

e) $\frac{x}{2} - \frac{4}{3} = \frac{2x-1}{3}$ f) $\frac{3x+2}{5} - \frac{3+x}{10} = \frac{2-x}{15} + \frac{1}{3}$

Gruble 3

a) Vis at

$$0{,}2626... = \frac{26}{99}$$

Gitt

$$a = b\left(\frac{1}{10^c} + \frac{1}{10^{2c}} + \frac{1}{10^{3c}} + ...\right)$$

hvor b er et tall med c siffer.

b) Vis at hvis $b = 26$, er $a = 0{,}2626...$.

c) Vis at

$$a = \frac{b}{10^c - 1}$$

Merk: Hvis for eksempel $a = 0.0909...$, må du regne $b = 09 = 9$ som et tall med 2 siffer.